



A novel Nb-W-C alloy with special microstructure and excellent mechanical property

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ABSTRACT

A new kind of Nb-W-C alloy with special microstructure and excellent mechanical property was designed and prepared by CVD subsequent with homogenizing heat treatment. Its microstructure is composed of columnar crystals with specific orientation, needle-like Nb₂C second phase and layered structure with Nb/W composition segregation. By analyzing the effect of homogenizing heat treatment on the property and microstructure of the alloy, it was found that the as-deposited layered structure was retained in the microstructure after heat treatment at 1400 °C, but disappeared by treated at 1600 °C. The layered structure can effectively separate Nb₂C and inhibit the size coarsening of Nb₂C during carbide formation. During the homogenization process, needle-like carbides can maintain the strengthening effect under the condition of slight grain growth, but excessively high homogenization heat treatment temperature coarsens the grains and carbides, and ultimately weakens the strength of the alloy. The comprehensive strengthening effect of grain refinement and carbide pinning makes the room temperature strength of the sample reach 670 MPa after homogenizing treatment at 1400 °C for 6 h, which is about 35–45% higher than the as-deposited Nb-W-C alloy and the existing widely used Nb521 alloy produced by melting-forging-annealing technology.

1. Introduction

Niobium (Nb) has the characteristics of low density (8.57 g / cm³), high melting point (2741 K), high plasticity, low vapor pressure and good corrosion resistance against liquid metal corrosion. Elevated temperature niobium-based alloy has good high temperature specific strength and processing performance, which can be used to make parts with complex shapes. It is one of the important candidate materials for aerospace structural parts [1–5]. Traditional niobium-based alloys have been used in aerospace and nuclear industries since the 1960 s [6], among which C-103 (Nb-10Hf-1Ti) alloy was the most widely used [7–9]. However, C-103 alloy is highly sensitive to oxygen, which easily leads to decrease in mechanical performance [10]. Since the 1990 s, with the development of space technology, higher requirements have been put forward for Nb-based high temperature structural materials. At

present, there are dozens of Nb alloys used in aerospace industry, which are mainly strengthened by adding W, Mo, Zr and Hf. The addition of W and Mo elements in the alloy can form a replacement solid solution because of their high melting point and similar atomic radius, so as to improve the room and high temperature strength of the alloy matrix. Nb521 (Nb-5 W-2Mo-1Zr) alloy, as a representative of Nb alloys with moderate strength and plasticity, its room temperature strength is 400–600 MPa, and its high temperature strength is higher at 1000–1400 °C [11–13].

And the addition of Zr and C can further improve the strength of the alloy, because C is easy to form a dispersed precipitation strengthening phase with Zr and Nb [14–16]. The strength of niobium alloy can be effectively improved by stabilizing fine carbides. The typical PWC-11 (Nb-1Zr-0.1 C) alloy is realized by adding Zr and C elements, its good high temperature creep resistance is related to carbides. Therefore, the

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carbide strengthening effect of Nb alloy has attracted much attention [17–20].

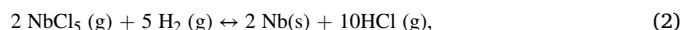
However, due to the extremely high melting point of refractory alloying elements, the traditional melting process of the refractory alloy is usually complex and difficult. Chemical vapor deposition (CVD) [21] has the characteristics of high purity, uniform density, controllable thickness, controllable crystal structure and relatively low cost. According to the temperature of CVD process is much lower than the melting point of the material, and the products are pure and free of impurities, it has great advantages compared with powder metallurgy and melting methods. Auck et al. [22] has successfully prepared W-22% Re alloy by CVD at different annealing temperatures and time. Wei et al. [23] studied the CVD of Nb on Mo substrate under different deposition variables, the deposited Nb has high purity and compact structure, and the grain size can be controlled by adjusting the deposition parameters.

Based on the above analysis of Nb-based structural alloys, refractory W and C were designed as alloying elements of Nb in this study, and CVD combined with homogenization heat treatment technology was adopted to produce a new kind of Nb-W-C alloy with excellent mechanical property. By characterizing the effect of microstructure evolution on mechanical properties of as-deposited and heat-treated alloys, the strengthening control factors of Nb-W-C alloy were clarified, and the microstructure optimization was guided to obtain performance improvement.

2. Materials and methods

The Nb-W-C alloy is deposited on Mo substrate via CVD. The raw materials used in this work are Nb plates ($\geq 99.99\%$), W plates ($\geq 99.99\%$) and molybdenum substrate (cylinder shape, size: $\Phi 30$ mm $\times$ 30 mm). The high purity carbon sheet is placed below the Mo substrate to provide carbon source. The schematic diagram for depositing Nb-W-C alloy on molybdenum substrate is presented in Fig. 1. The chlorination chamber and the deposition chamber were heated by electric furnace and medium frequency induction furnace respectively. The pressure of the deposition chamber is controlled by the vacuum pumps, and the flow rates of Cl_2 and H_2 are independently controlled by the mass flow controllers. The deposition temperature (T_d) is fixed at the range of 1100°C – 1400°C , where the chlorination temperature (T_c) is 750°C , chlorine flow (F_{Cl_2}) is in the range of 50–300 ml/min, hydrogen flow (F_{H_2}) is in the range of 300–800 ml/min, and deposition time is 48 h. The chlorination temperature of niobium is 300°C , while that of tungsten is above 700°C . At high temperatures, gaseous NbCl_5 is formed

by in-situ chlorination shown in Eq. (1). After the sublimation of NbCl_5 , the precursor is adsorbed on the high temperature substrate and reacts with hydrogen to form niobium atoms, as shown in Eq. (2). The by-product HCl gas is treated by neutralization process. While the formation of WCl_6 and the reduction of W are similar to this process (Eq. 3–4). Carbon can be vaporized at 1000°C , and the C atoms in the deposition process can combine with Nb atoms to form Nb_2C (Eq. 5). The chemical composition of the Nb-W-C alloy was obtained by chemical analysis method, as shown in Table 1. The original deposition thickness of CVD samples on the substrate is 4 mm. The test samples in this study were taken from the middle of the deposited bulk. The upper and lower parts of the deposited bulk were cut off by 1 mm thickness, leaving the middle.



According to the size requirements of Fig. 3, the as-deposited bulk were processed into tensile samples by wire cutting, leaving enough samples for homogenization heat treatment. And the sample size for room-temperature tensile test is shown in Fig. 3. The thickness of the samples is 2 mm. The process diagram of the homogenized heat treatment sample is shown in Fig. 2. The samples after deposition were homogenized at 1400°C and 1600°C for 4 h or 6 h, respectively. The samples were homogenized in a high temperature vacuum furnace. The sample was placed in a vacuum furnace and heated with the furnace. The heating rate was $10^\circ\text{C} / \text{min}$, and the sample cooled with furnace after heat treatment. Before the tensile test, the surface of samples was polished with 400 # and 1200 # sandpaper, so that the surface smoothness can meet the test requirements. Sample pretreatment can eliminate the influence of surface defects on the mechanical properties. Then the mechanical properties of the samples in different states were tested.

Uniaxial tensile tests at room temperature were carried out on Shimadzu Tester Testing machine. The samples that as-deposited and completed homogenizing treatments were ground by sandpaper along the radial direction, and then polished. The microstructure was characterized by optical microscopy (OM, LEICA DM4000M), scanning electron microscope (FE-SEM, FEI VERSA 3D) and electron back-scatter diffraction (EBSD). The composition segregation of Nb and W elements layered structure was observed by electron probe X-ray micro-analyser (EPMA, JEOL JXA8230). Particle size distribution software was used to calculate the grain size of the samples. The size and area fractions of the second phase were quantitatively analyzed by an automatic image analysis software (Image-Pro Plus 6.0). Besides, the type of the second phase was determined by selected area electron diffraction (SAED) technology by using a transmission electron microscopy at 300 kV (TEM, FEI TECNAI G² F30). The foils for the TEM observation were prepared first by hand grinding to 50 μm in thickness and then thinned using an electro-polisher with twin-jet in 4% HClO_4 solution in ethanol as electrolyte.

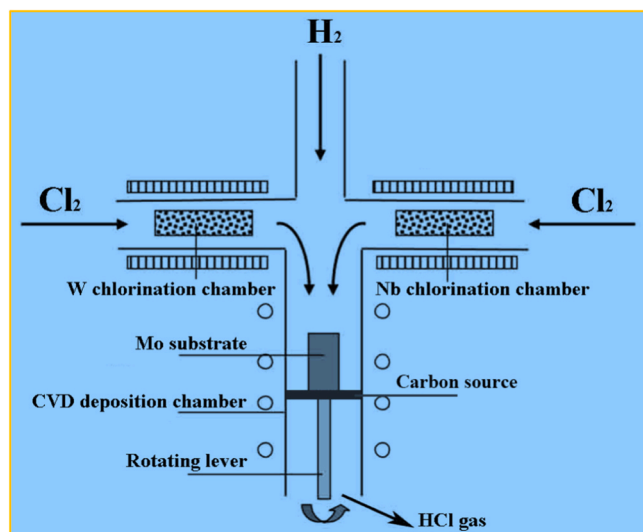


Fig. 1. The schematic diagram of CVD Nb-W-C alloy.

Table 1

Chemical composition of the experimental Nb-W-C alloy (mass fraction, %).

Element	Nb	W	C
Contents	Bal.	10 ± 0.08	2.5 ± 0.02

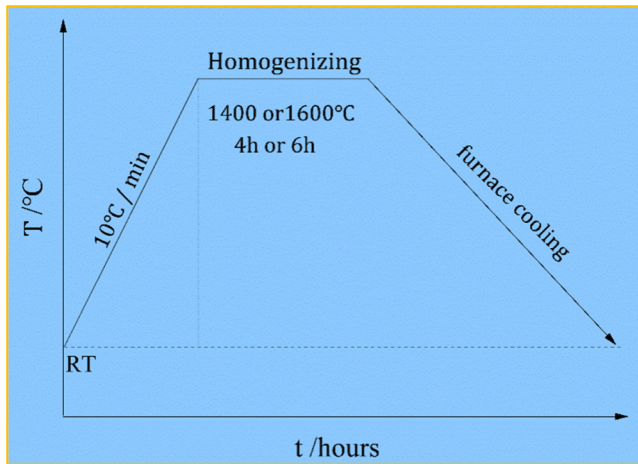


Fig. 2. Homogenization treatment process diagram.

3. Results and Discussion

3.1. Microstructure evolution

Fig. 4 illustrates typical metallography of the samples as-deposited and homogenizing-treated at different temperatures. In this figure, microstructure of the as-deposited sample and the samples treated at 1400 °C consists of columnar grains, layered structure and second phase. As shown in metallography images, the dark gray structure is second phase, the white bright structure is (Nb,W)_{ss} grain. The layered structure runs through Nb-W grains and is perpendicular to the deposition direction. When the homogenizing temperature increases from 1400 °C to 1600 °C (Fig. 3b to e), The grain size grows rapidly, and the morphology and distribution of the second phase also change significantly. With the increase of homogenization temperature and holding time, the layered structure gradually fades. The second phase in the microstructure gradually grow from needle-like to long rod-like with the increase of homogenization temperature and holding time. At 1600 °C, the layered structure disappears. The higher temperature contributes to the redistribution and homogenization of Nb and W elements, making the composition gradient in the layered structure disappear. And the number and size of the second phase particles increases significantly. This is because after holding at a higher temperature, the elements are fully dissolved in the grains and promote precipitation during the subsequent slow cooling process.

Fig. 5 shows the electron channeling contrast imaging (ECCI) morphology of Nb-W-C alloy obtained by different heat treatments. The ECCI technique can not only obtain the atomic number contrast image (Z-contrast), but also provide the information of grain orientation difference, which can better characterize the microstructure characteristics. In recent years, it has been well applied in metal material characterization [24,25].

Fig. 5 clearly shows that the grain size, grain orientation and layered composition gradient structure have changed significantly after

different heat treatments. Under the same accelerating voltage, with the increase of solution temperature and holding time, the composition gradient structure in ECCI images disappears gradually. At the same time, the grain size gradually increases, accompanied by grain orientation transformation. The length and width of columnar grains in different states were analyzed in Table 2.

By comparing the average grain size, it is found that after heat treatment at 1600 °C, the grain size increases to more than 20 times of the as-deposited alloy. Compared with the as-deposited alloy, homogenization temperature and holding time can promote grain coarsening. With the increase of temperature and holding time, the length-width ratio of columnar grains decreases significantly. From the initial $L/W = 3.283$ to $L/W = 1.215$, the columnar grains gradually changed to equi-axis. It is well known that the driving force for grain growth is the energy associated with the decrease in grain boundary area. The driving force of grain boundary movement is driven by Laplace square [26]:

$$P = \gamma \left(\frac{1}{\rho_1} - \frac{1}{\rho_2} \right) \quad (1)$$

where γ is the interface energy; ρ_1 and ρ_2 are two curvature radius of adjacent grain boundaries, respectively. When the grains with radius R_A are in contact with the surrounding grains, the driving force between the two grains is:

$$P_{AB} = 2\gamma \left(\frac{1}{R_A} - \frac{1}{R_B} \right) \quad (2)$$

where P_{AB} is the driving force of grain B to grain A; γ is the interface of grain A and grain B. When $R_A < R_B$, grain A grows up by swallowing grain B.

During the deposition process, the layered structure with component segregation is formed layer by layer, the deposition process accumulates large stress inside the grains. And the second phase is formed at the columnar grain boundary, the interface energy between the grains is larger. Therefore, the grain boundary in the long axis direction of the columnar crystal has higher interface energy (γ). Thus, the grain boundary in the long axis direction is rapidly migrated by heating during the subsequent homogenization process. The migration rate of the long axis direction of the columnar crystal is faster, so as the temperature increases, the grains gradually become equiaxed. In addition, the large grains gradually swallow up the small grains through the migration of grain boundaries, and the grains are coarsened.

3.2. The layered structure

Fig. 6 shows the composition segregation of Nb and W elements in the as-deposited sample. The BSE image shown in Fig. 6(a) has a clear Z-contrast, and the W element with a large atomic number is bright. After CVD deposition, Nb-W solid solutions composed of continuous regular layered structure with composition segregation was obtained.

Based on thermodynamic analysis and microstructure observation, it is found that the chlorination reaction of Nb and W is a competitive reaction type. The formation mechanism of the layered structure can be expressed as Fig. 7. Through the analysis of the EPMA scanning of the

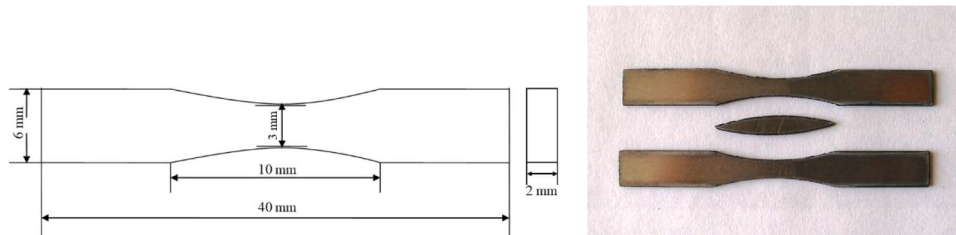


Fig. 3. Sample size and actual appearance for room-temperature tensile test.

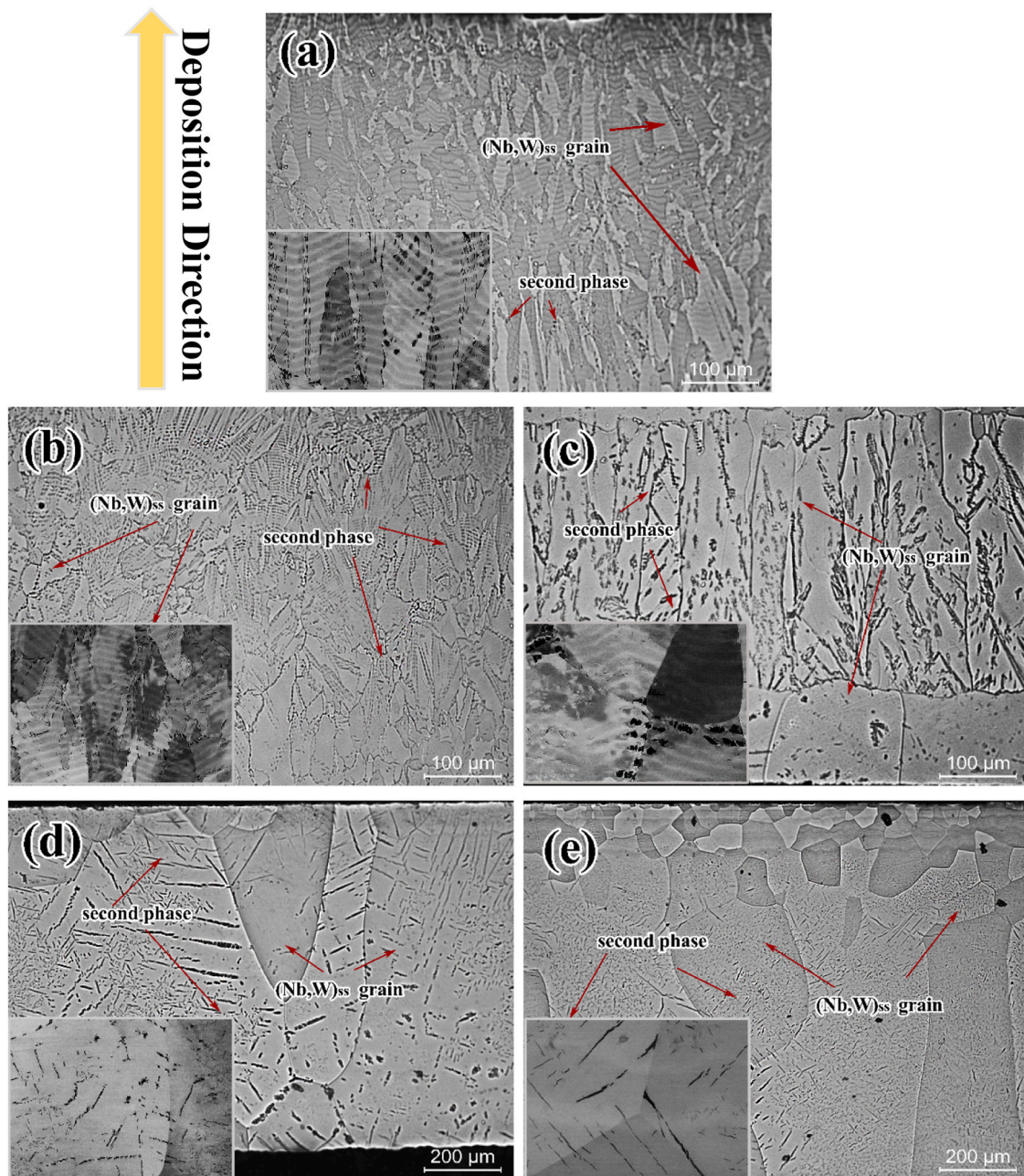


Fig. 4. Metallography of the CVD Nb-W alloy homogenizing by different processes (a) as-deposited, (b) 1400 °C-4 h, (c) 1400 °C-6 h, (d) 1600 °C-4 h and (e) 1600 °C-6 h.

layered structure (Fig. 6), it is inferred that the chloride of W was adsorbed on the surface of the substrate during CVD deposition, and then reduced by H_2 to form a W segregation layer. After a period of reaction, there is enrichment of niobium chloride near the W segregated layer. Under the action of diffusion driving force caused by concentration difference, the reaction active center begins to transform into the adsorption of niobium chloride. After that, H_2 reduced niobium chloride to Nb atoms. The presence of Nb atoms provides conditions for further adsorption of niobium chloride and promotes the formation of Nb segregation layer. The final result of the above process is the formation of Nb-W layered structure. The final result of the cyclic alternation of the above processes is the formation of Nb-W layered structure with composition segregation.

As shown in Fig. 8, the layered structure is arranged layer by layer along the crystal growth direction and can still be connected at the

boundary. And the layers are arranged closely and naturally connected at the interface of adjacent columnar grains. The average thickness of the layered structure in the sample is between 5 and 15 μm . According to the atomic number contrast, the bright layer is W-rich and the dark layer is Nb-rich. Based on the above characterization, it is found that the sample after heat treatment at 1400 °C still maintains a layered structure. The results of EDS spot analysis (Fig. 8a-c) show that after heat-treated at 1400 °C, the W content in the layered structure decreases and the Nb content increases with longer holding time. After heat-treated at 1600 °C, the structure of Nb-rich and W-rich segregation layer disappears and the composition in the alloy tends to be uniform (Fig. 5d-e).

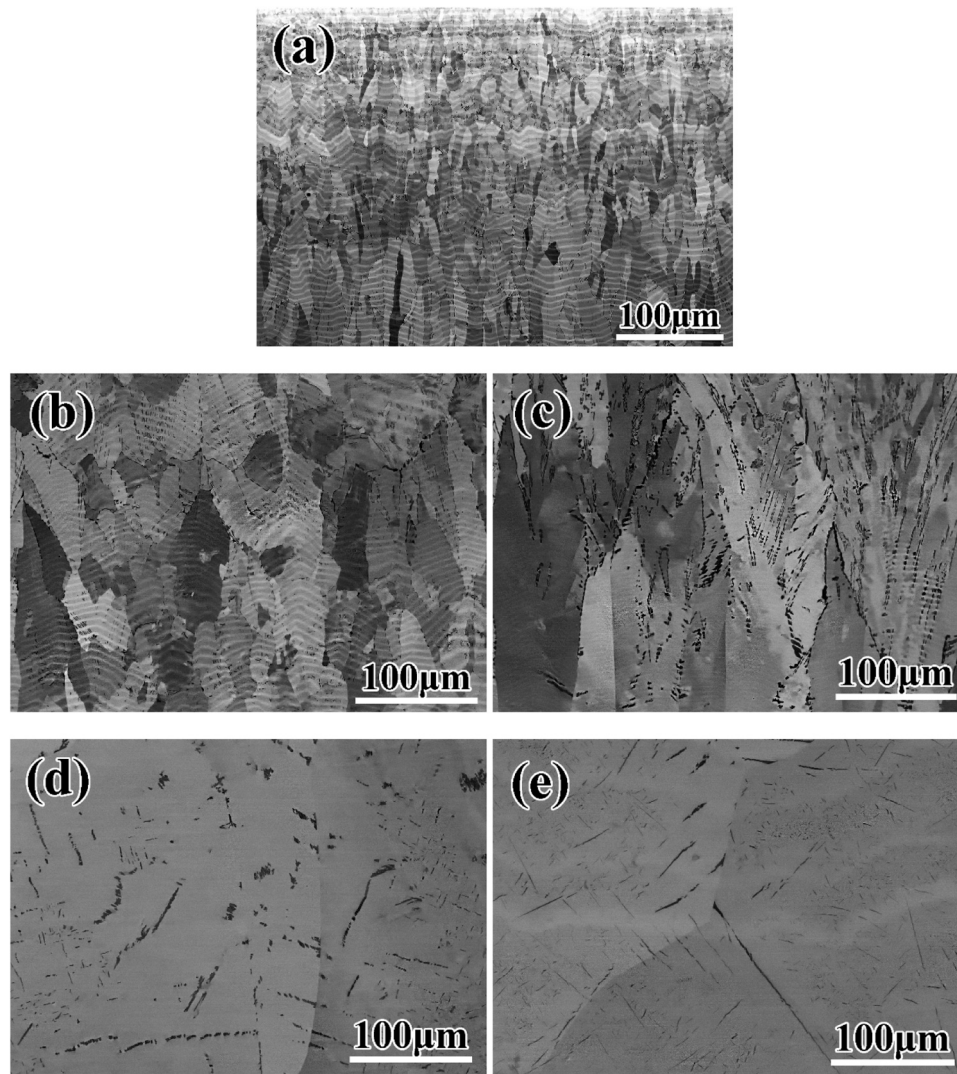


Fig. 5. ECCI images of the CVD Nb-W-C alloy heat treated by different processes (a) as-deposited, (b) 1400 °C-4 h, (c) 1400 °C-6 h, (d) 1600 °C-4 h and (e) 1600 °C-6 h.

Table 2

Grain comparison of the Nb-W-C alloy under different treatment conditions.

Treatments	As-deposited	1400 °C-4 h	1400 °C-6 h	1600 °C-4 h	1600 °C-6 h
Average Grain Size D(μm)	9.67	19.22	38.26	66.89	214.26
Average Length(μm)	15.53	21.63	41.91	68.41	209.32
Average Width(μm)	4.73	13.41	27.43	51.37	172.26
Grain Aspect Ratio (L/W)	3.283	1.613	1.528	1.332	1.215

3.3. The second phase

Fig. 9 shows the selected area electron diffraction (SAED) and bright field image (BF) of the matrix and the second phase observed from the as-deposited sample. The electron diffraction patterns of the matrix obtained in three low-index crystal band of [001], [011] and [111] axis directions are calibrated in Fig. 9(a). The diffraction information conforms to the electron diffraction pattern characteristics of bcc structure. The calculated lattice constant of the matrix is $a = 3.256 \text{ \AA}$, which is between Nb ($a = 3.294 \text{ \AA}$) and W ($a = 3.191 \text{ \AA}$). It indicates that the

matrix is $(\text{Nb,W})_{\text{ss}}$. From the bright field image of Fig. 9(b), it can be clearly seen that second phases are discontinuously distributed on the grain boundaries. SAED patterns of [0001] and [10–10] hexagonal axes are shown in Fig. 9(c), which represent as typical electron diffraction patterns of hexagonal structure. Since the crystal structure of $\alpha\text{-Nb}_2\text{C}$ is an orthorhombic structure, which is inconsistent with the SAED diffraction results, the possibility is excluded. The ratio of lattice constant a/c of $\beta\text{-Nb}_2\text{C}$ is 1.088, while the ratio of lattice constant a/c of $\gamma\text{-Nb}_2\text{C}$ is 0.629. Through the diffraction patterns calibration, the lattice constants of Nb_2C were calculated. The calculated lattice constants of this second phase are $a = 5.44 \text{ \AA}$, $c = 5.0 \text{ \AA}$, $a/c = 1.088$. Therefore, it is inferred that the second phase is $\beta\text{-Nb}_2\text{C}$.

The STEM-EDS of the as-deposited sample is shown in Fig. 10, which gives the chemical composition information of the matrix and the precipitated phase. As can be seen from Fig. 10(a) and (b), the matrix contains Nb and W elements, and the second phase contains Nb and C elements. It can be clearly seen from the mapping scanning results in Fig. 10 (c)-(f) that the second phase is poor in W, and rich in Nb and C. From Fig. 11 (a) and (b), it is found that Nb_2C is obviously distributed in the darker contrast layer which is rich in Nb with smaller atomic weight. Among them, some carbides are located inside the grains, and some carbides are located at the grain boundaries, but all carbides are basically concentrated in the Nb-rich layer. The black rod-like particles

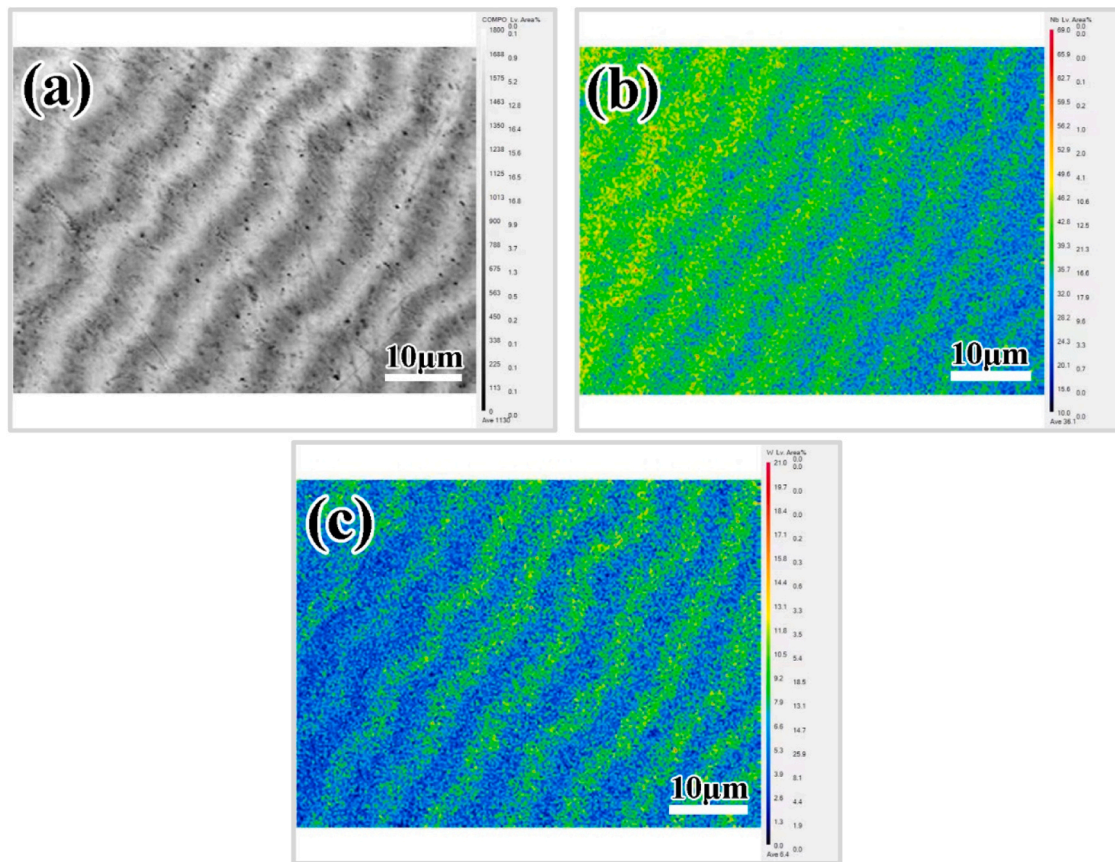


Fig. 6. Composition segregation of Nb and W elements in EMPA mapping of CVD Nb-W alloy (a) BSE image, (b) composition segregation of Nb and (c) composition segregation of W.

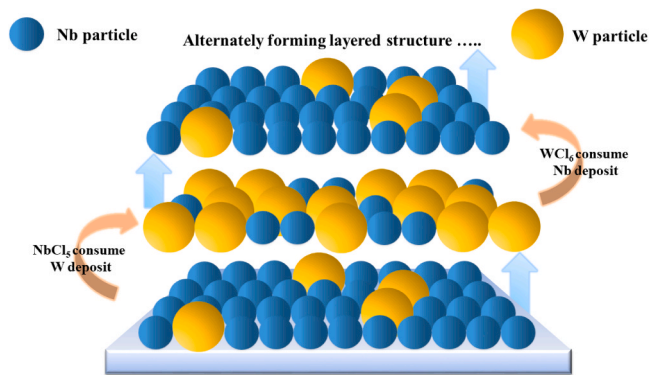


Fig. 7. The formation mechanism of Nb / W layered structure.

shown in the figures are Nb_2C . It is observed from Fig. 11 that, there are discontinuous Nb_2C separated by W-rich layers at grain boundaries in the as-deposited sample. The carbides are small in size and have an ellipsoidal or short rod shape. After high temperature heat treatment, the Nb-W segregation layered structure disappears, and the carbides gradually merge and grow along the long axis of the particles. The length distribution and area proportion of Nb_2C under three treatment conditions were compared and statistically analyzed. The results are shown in Fig. 12 and Table 3.

The carbide length distribution range and frequency in different treatments are shown in Fig. 10. It is found that the length of Nb_2C in the as-deposited sample is mostly less than 2 μm , and more than half of these short rod carbides have a length of less than 0.5 μm . After homogenizing treatment at 1400 $^{\circ}\text{C}$ for 4 h, the length of Nb_2C is concentrated in the

range of 1–5 μm . Driven by thermal energy, the fine Nb_2C grows to a size of more than 0.5 μm . After treated at higher temperature of 1600 $^{\circ}\text{C}$, Nb_2C coarsens obviously and the number of Nb_2C with length more than 5 μm increases sharply. At the same time, small-sized Nb_2C further merge and grow in the length direction. The coarsening of brittle second phase increases the weak links inside the grains. In the process of bearing load, it is easier to produce stress concentration cracking, which leads to the decrease of mechanical properties.

By comparing Table 3, it is found that the area fraction of Nb_2C increases significantly after heated by high temperature. Although the area fraction of Nb_2C is similar at 1400 $^{\circ}\text{C}$ and 1600 $^{\circ}\text{C}$, the finer carbides at 1400 $^{\circ}\text{C}$ strengthen the alloy, while the coarse carbides at 1600 $^{\circ}\text{C}$ deteriorate the alloy performance. From the SEM-EDS analysis of Nb_2C carbides, it is found that the Nb content in carbides increases, the W and C contents decrease after homogenizing treatment. As the temperature further increases, the C content further decreases. It is concluded that high temperature makes C element more fully dissolved in the matrix, and high temperature heat treatment is beneficial to the improvement of alloy performance to a certain extent. Due to the electron beam spot diameter of SEM-EDS is much larger than that of STEM-EDS, the influence of matrix composition is often brought into the test results. This is the reason why the Nb_2C composition in the table is different from the STEM-EDS results.

3.4. Grain orientation

Fig. 13 shows the IPF mapping and IPF distribution collected along the deposition direction of the sample under different heat treatment conditions. The inverse pole figure in Y_0 direction represents the distribution density of grains in three different orientations along the deposition direction. The grains in the as-deposited sample are fine, and

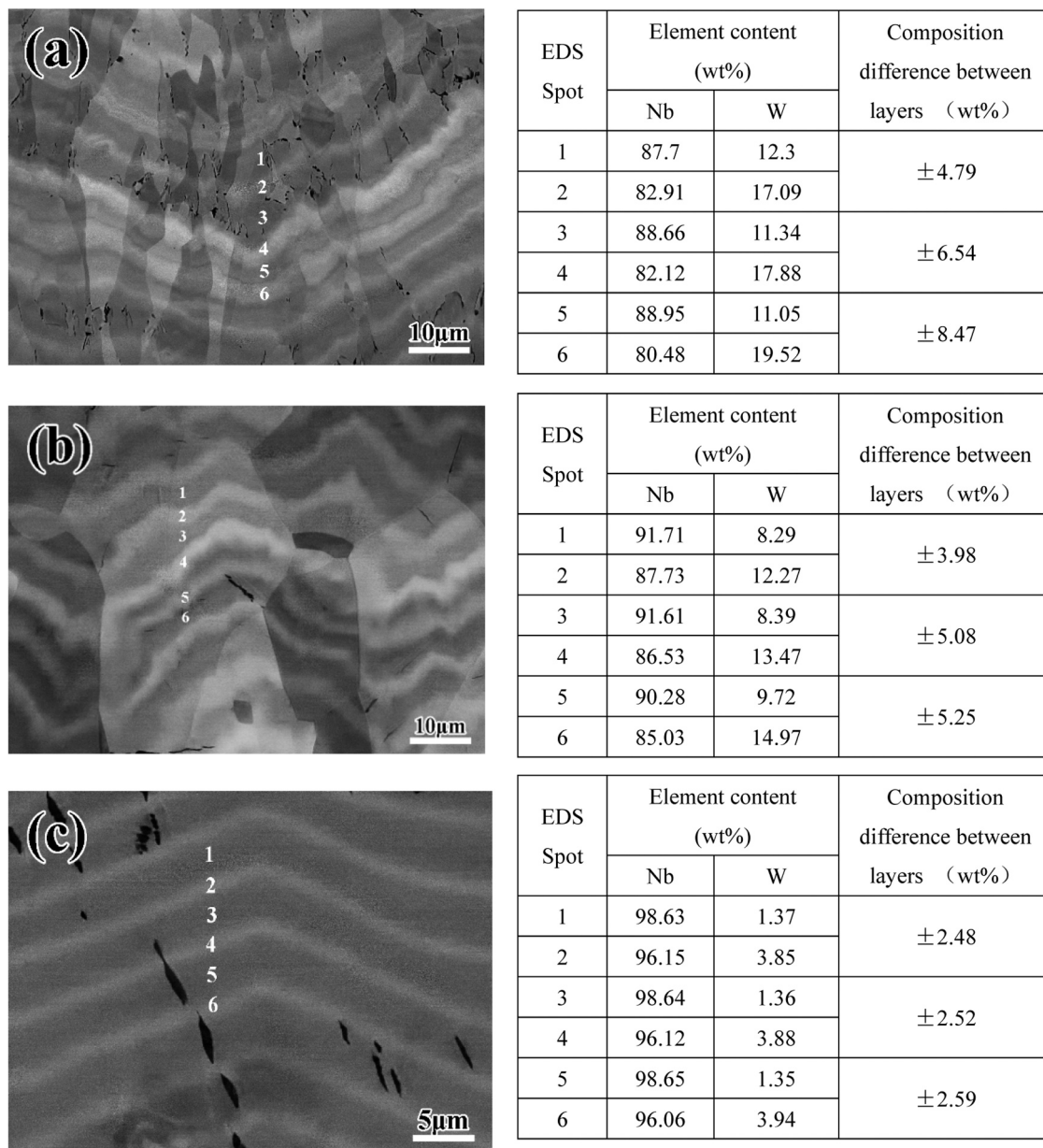


Fig. 8. EDS spot analysis of the CVD Nb-W alloy heat treated by different processes (a) as-deposited, (b) 1400 °C-4 h, (c) 1400 °C-6 h.

most of them are mainly inclined to two orientations of $\langle 100 \rangle$ and $\langle 111 \rangle$. After heat-treated at 1400 °C, the grains grow into columnar crystals along the deposition direction, and concentrated in the preferred orientation of $\langle 100 \rangle$. After heat-treated at 1600 °C, the grains transform to equiaxed grains, and the preferred orientation of grains tends to be $\langle 111 \rangle$. In general, after high temperature heat treatment, the as-deposited grain morphology changes from columnar crystal to equiaxed crystal. Since close-packed direction of the BCC is $\langle 111 \rangle$, according to the principle of the lowest free energy of the system, the preferentially $\langle 001 \rangle$ oriented grains gradually rotate toward the orientation of $\langle 111 \rangle$.

3.5. Tensile behavior after homogenizing treatment

The microstructure of the deposited alloy changed significantly after high temperature heat treatment. At 1400 °C, the grain size can still be consistent with the as-deposited grain size, and the grains grow rapidly at 1600 °C. The engineering stress-strain curves of the tested samples are compared in Fig. 14(a), and their corresponding tensile properties

are summarized in Fig. 14(b).

Compared with the as-deposited alloy, the stress-strain curves of the alloy after homogenizing treatment show completely different mechanical behavior. According to the dates, the strength of the sample reach 670 MPa after homogenizing treatment at 1400 °C for 6 h, which is about 35–45% higher than the as-deposited Nb-W-C alloy and the existing widely used Nb521 alloy produced by melting – forging – annealing technology, and the elastic modulus (E, ratio of stress to strain) of the homogenizing treated alloy is slightly higher than that of the as-deposited alloy. As the solution temperature increases from 1400 °C to 1600 °C, the tensile strength of the alloy decreases by about 95% after holding for 6 h, as shown in Fig. 14(b).

The tensile strength decreased significantly at 1600 °C due to the grain coarsening caused by the increase of homogenizing temperature. Grain coarsening results in embrittlement of the alloy, which reduces tensile strength seriously. The recrystallization temperature of the as-deposited alloy is estimated to be lower than 1400 °C, according to 0.4 times of the melting point of the alloy. Therefore, the Nb-W-C alloy in this work is prone to recrystallization at treatment temperatures. In

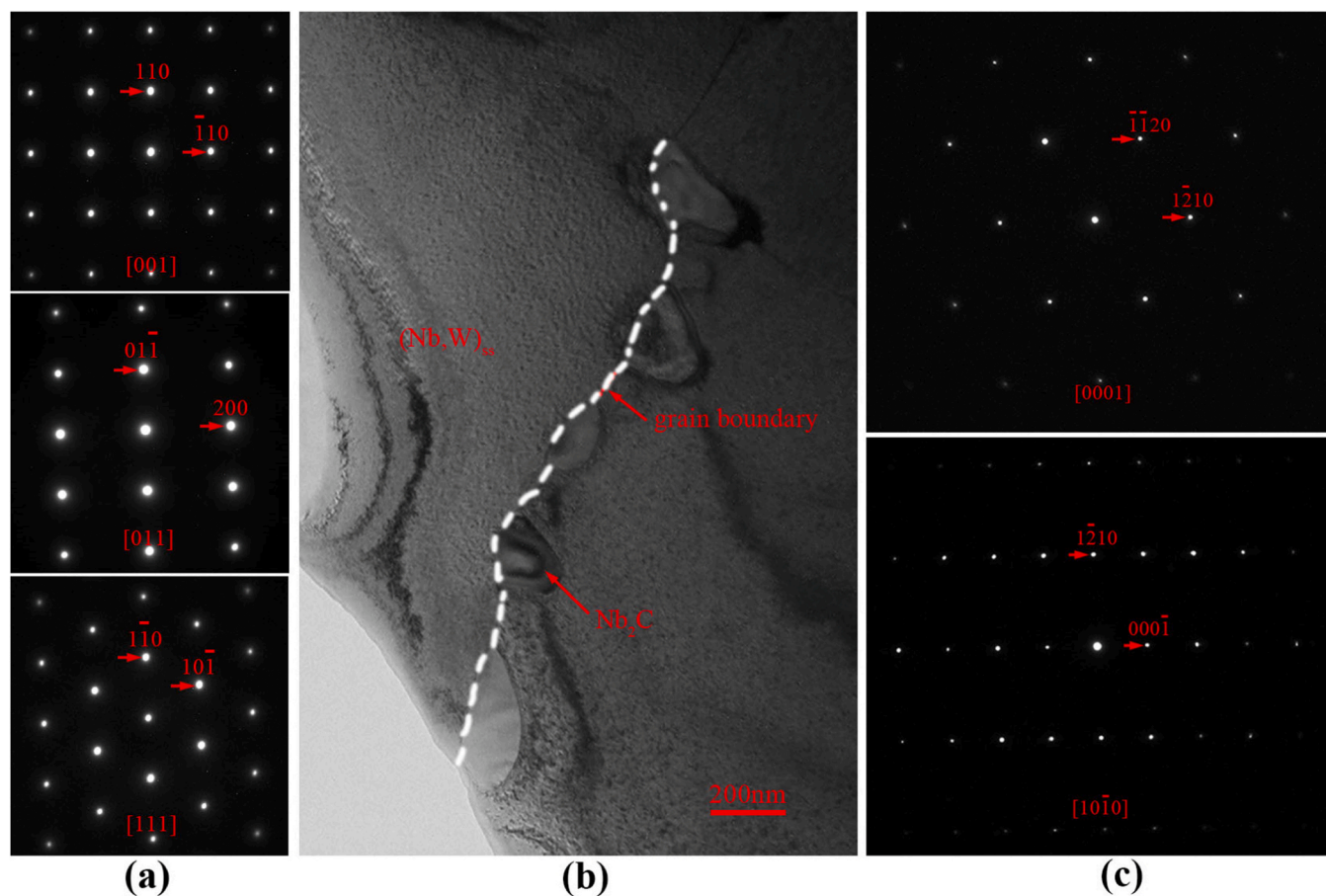


Fig. 9. Bright field image and selected area electron diffraction patterns of $(\text{Nb,W})_{\text{ss}}$ and second phase at the matrix grain boundary. (a) $(\text{Nb,W})_{\text{ss}}$ characteristic band-axis diffraction patterns of $(\text{Nb,W})_{\text{ss}}$, (b) BF image and (c) characteristic band-axis diffraction patterns of Nb₂C patterns.

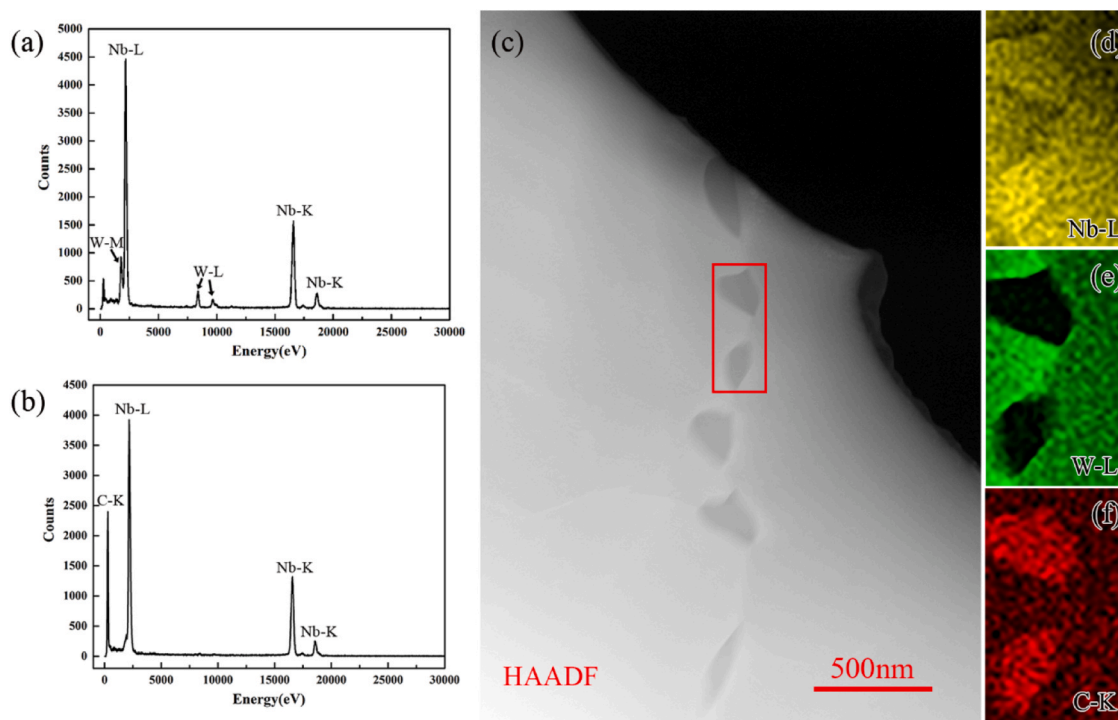


Fig. 10. STEM-HAADF image and EDS analysis of the $(\text{Nb,W})_{\text{ss}}$ and Nb₂C (a) STEM-EDS of $(\text{Nb,W})_{\text{ss}}$, (b) STEM-EDS of Nb₂C, (c)-(f) STEM-HAADF image and EDS mapping of Nb-L, W-L and C-K.

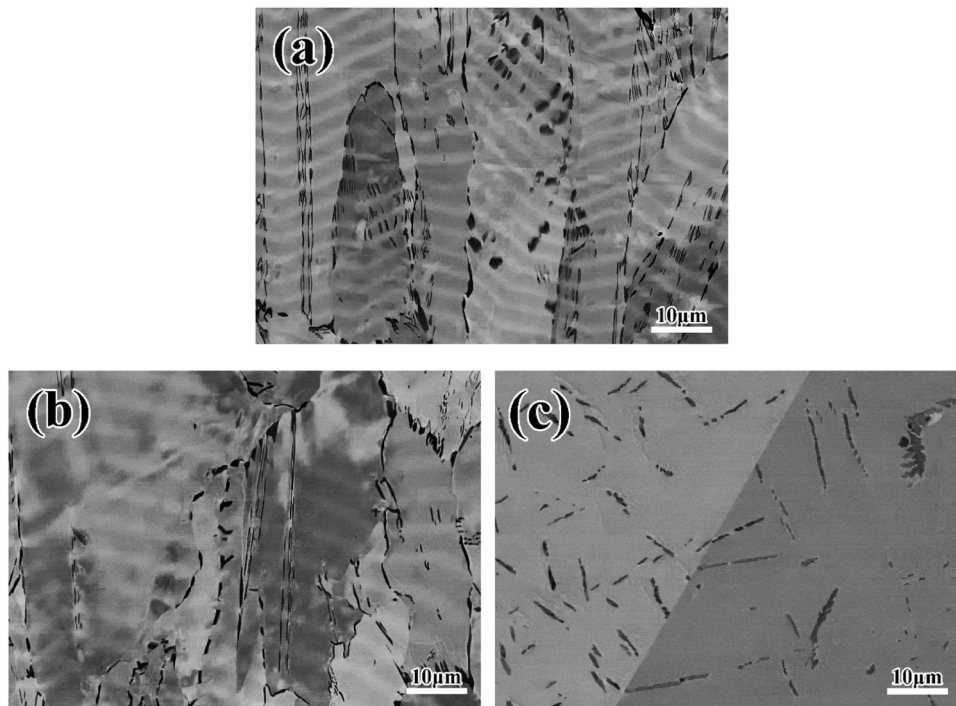


Fig. 11. Nb₂C evolution in CVD Nb-W-C alloy treated by different processes (a) as-deposited, (b) 1400 °C-4 h, (c) 1600 °C-4 h.

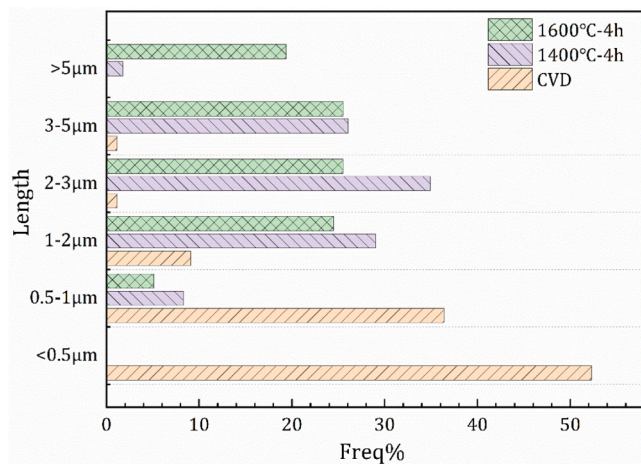


Fig. 12. Relationship between Nb₂C length distribution and frequency in different treatments.

Table 3
SEM-EDS analysis and area fraction of Nb₂C with different treatments.

Nb ₂ C	Area fraction%	Nb (wt%)	W (wt%)	C (wt%)
As-deposited	1.385	70.21	11.16	18.63
1400-4 h	3.608	81.21	4.46	14.34
1600-4 h	3.65	84.7	5.93	9.37

summary, homogenizing of alloying elements, grain refinement and carbide dispersion strengthening are important strengthening methods for tested alloy. When the precipitation of fine Nb₂C increases, the strength of the alloy is significantly improved. When Nb₂C is combined and coarsened, the grains coarsen sharply, resulting in a decrease in the mechanical properties of the alloy. The amount of precipitation strengthening was estimated by Ashby-Orowan [27] mechanism as follow:

$$\Delta\sigma = \frac{5.9\sqrt{f}}{d} \left(\ln \frac{d}{2.5 \times 10^{-4}} \right) \quad (3)$$

where d is the average particle size of the second phase, μm ; f is the volume fraction of second phase; σ is the contribution of the second phase to the yield strength, MPa. Combined with the analysis of Nb₂C size and area fraction in Table 3 and Fig. 12, it can be seen that carbides with a small size are beneficial to the improvement of strength, and the increase of carbide volume fraction can also improve the strength. Compared to the as-deposited alloy, the area fraction of Nb₂C increased by nearly 3 times after homogenization treatment, which strengthened the alloy to a certain extent. However, according to the Hall-Petch relationship, grain refinement also plays a great role. According to the above analysis in Table 2, after homogenization treatment at 1600 °C, the grain size of the alloy increases by nearly 20 times, so the properties of the alloy are greatly weakened, and the strengthening effect of carbides will not be significantly reflected. Although the grain size was coarsened to a certain extent after homogenization treatment at 1400 °C, the strength of the alloy was improved due to the increase of carbide precipitation.

The tensile fracture surfaces under room temperature are shown in Fig. 15. From Fig. 15(a)-(d), the fracture morphology has changed from cleavage to intergranular fracture. The cleavage fracture shows layered characteristics, which is consistent with the structure mentioned above. It is also verified that the increase of homogenizing temperature effectively promotes grain coarsening. The grain size of the 1400 °C sample is much smaller than that at 1600 °C, the fine grain is beneficial to the strength improvement. In the sample solution treated at 1600 °C, due to the high hardness, brittleness and poor deformability of Nb₂C, cracks can easily initiate at the interface between carbides and (Nb,W)_{ss} matrix during tensile deformation. The larger Nb₂C at the grain boundary leads to a decrease in the interfacial bonding force during the tensile process, so the strength becomes lower.

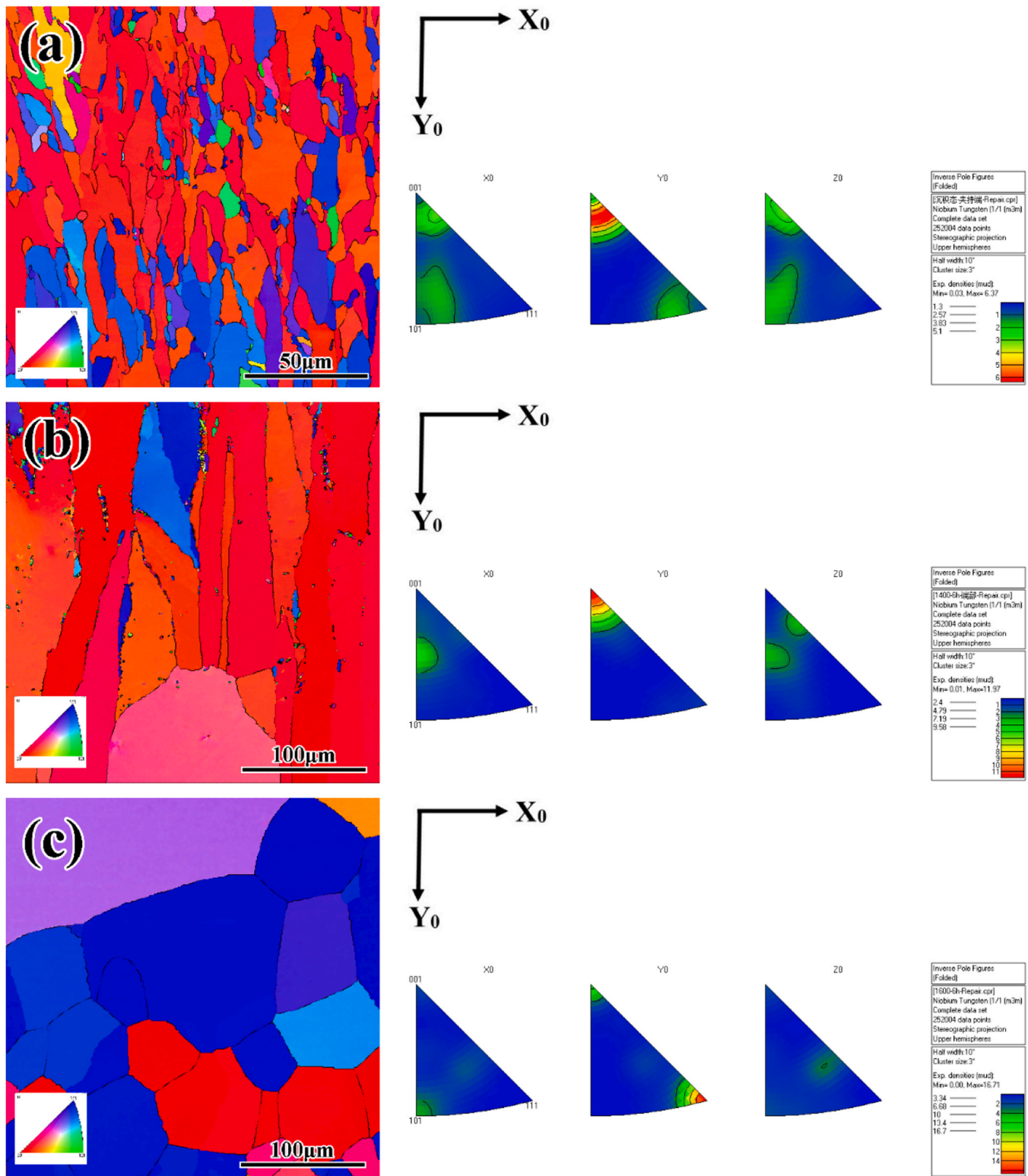


Fig. 13. IPF mapping and IPF distribution along the deposition direction of the Nb-W alloy heat treated by different process (a) as-deposited, (b) 1400 °C-6 h, (c) 1600 °C-6 h.

4. Conclusions

(1) A new kind of Nb-W-C alloy with special structure was prepared by CVD combined with homogenizing heat treatment. It is quite different from the alloy structure obtained by melting method. The microstructure is composed of columnar crystals with

specific orientation, Nb₂C second phase and layered structure with composition segregation of Nb/W.

(2) The layered structure can effectively separate Nb₂C and inhibit the size coarsening of Nb₂C during carbide formation. As the heat treatment temperature increases to 1600 °C, Nb₂C gradually

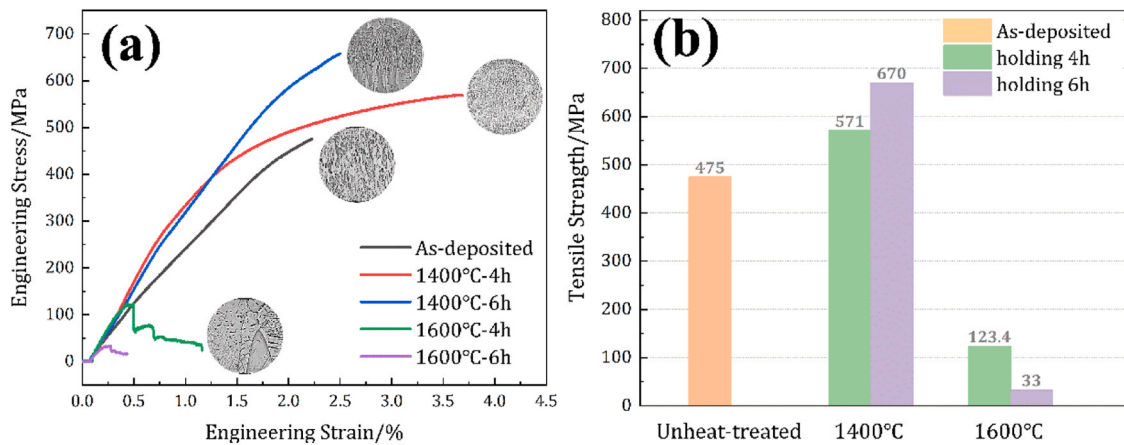


Fig. 14. Typical tensile engineering stress-strain curves of the tested samples.

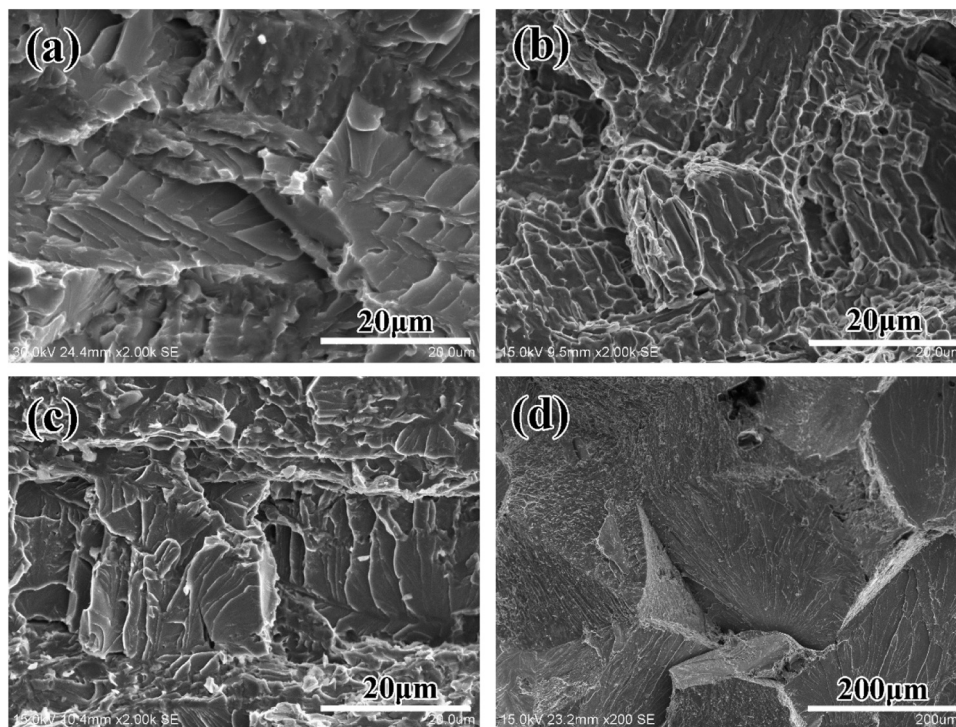


Fig. 15. The SEM images of tensile fracture surfaces for samples treated (a) none, (b) at 1400 °C for 4 h, (c) at 1400 °C for 6 h, and (d) at 1600 °C for 4 h.

merges and grows due to the disappearance of the layered structure.

- (3) The tensile strength of the sample reach 670 MPa after homogenizing treatment at 1400 °C for 6 h, which is about 35–45% higher than the as-deposited Nb-W-C alloy and the existing widely used Nb521 alloy produced by melting-forging-annealing technology. The excessively high heat treatment temperature coarsens the grains and carbides, and finally makes the fracture mode of the alloy change from cleavage fracture to intergranular fracture.

CRediT authorship contribution statement

Xiaohong Yuan: Conceptualization, Methodology, Investigation, Formal Analysis, Writing-Original Draft, Review & Editing; Yan Wei & Yong Han: Conceptualization, Funding Acquisition, Resources, Supervision, Writing-Review & Editing; Wanchun Zhao & Li Chen: Funding

Acquisition, Supervision; Xian Wang: Validation and Investigation; Changyi Hu & Jialin Chen: Supervision; Shanju Zheng: Writing-Review & Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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